Learning Patterns-Guided Data Generation for Knowledge Tracing

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Abstract

Knowledge tracing (KT), which is instrumental in monitoring and forecasting students' knowledge states throughout their learning trajectory in online learning environments, has over the past decade garnered widespread attention due to its pivotal role in facilitating personalized education. Existing KT approaches were mainly invented from the model-centric perspective to overcome the sequence modeling difficulty while not exploiting the potential information of sparsity, thereby limiting their performance. To make full use of the information in the dataset, this paper proposes a data-centric knowledge tracing paradigm, termed **LPDG**, aiming to generate interaction data between students and exercises by revealing students' **L**earning **P**atterns and facilitating the **G**eneration of ideal training **D**ata. Specifically, we propose a learning patterns-guided exercise sequence regenerator, which incorporates Transformer and a tailor-made pattern enhancer, thereby aiding in the extraction of valuable information for generating high-quality training data. Moreover, we devise a learning patterns-guided pseudo-label generator, which leverages the diffusion process to construct pseudo-labels for the regenerated sequences. Afterwards, the fully generated ideal data is incorporated into the training data, and we integrate this framework with various model-centric approaches in KT. Finally, experimental results across datasets clearly demonstrate the efficacy of our proposed LPDG framework.

CCS Concepts

• **Applied computing** → **E-learning**.

Keywords

Knowledge Tracing,Data Generation,Diffusion Model,Learning Patterns

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1 Introduction

With the rapid development of online education, the progress of intelligent tutoring systems has promoted the rise of a new paradigm of data-driven computer-assisted education. Knowledge Tracing, a focal point of research within this domain, aims to analyze students'

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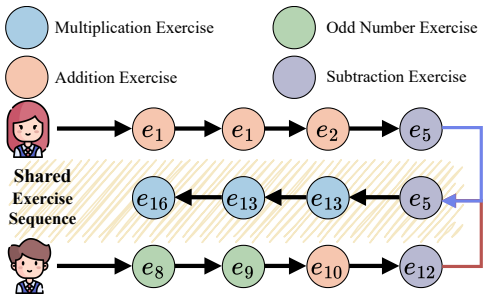


Figure 1: The figure reveals that although the two students started with different exercises, there is a common subsequence in their answer sequences.

exercise data to monitor real-time changes in knowledge states, predict future learning performance, and thereby provide personalized learning services and tailored exercise recommendations.

The primary challenge in knowledge tracing lies in accurately modeling the sequence of interactions between students and concepts to precisely capture their knowledge state. In recent years, the rapid development of deep learning has propelled the application of temporal models such as Recurrent Neural Networks(RNN), Long Short-Term Memory(LSTM) [9], and Transformers [30], yielding significant results. However, existing methods often involve designing complex models to enhance performance. For instance, stableKT [14] enhances knowledge tracing models' length generalization through linear bias and employs parallel Euclidean and hyperbolic attention heads, and finally aggregates these representations through a projection layer before the sigmoid prediction head. QRCL [46] improves the exercise response representation in knowledge tracing in a self-supervised manner by utilizing three views (the original interaction view, the contrastive enhancement view, and the collaborative response relationship view) and combining two-layer contrastive learning (the interaction layer and the structure layer), thereby enhancing the accuracy of predicting students' future performance.

Nevertheless, this capturing students' knowledge states is dataset-dependent. When the dataset exhibits excessive sparsity, model may be overfitting, as the quality of data fundamentally limits model performance [12, 44, 47]. If learning exercise sequence and student's response paths lack diversity or consist of repetitive patterns, model-centric approaches lose their foundation, resulting in knowledge states that lack generality and are prone to overfitting. This not only hinders understanding of student learning dynamics but also limits applicability of knowledge tracing models.

Therefore, a more promising solution focuses on the data-centric paradigm, which involves uncovering the intrinsic universality within student interaction data to provide informative support for complex model-centric paradigms. For example, in the field of cognitive diagnosis, CMES [22] has proposed a high-quality data sampling method centered on data to enhance the existing cognitive diagnosis models. To explore these implicit and generalizable interaction data, we analyzed real-world datasets, revealing similar learning behaviors among students. This similarity can be attributed to the concepts that govern feature acquisition, requiring not only common learning pathways but also reliance on the inherent logical relationships among concepts. We define these similar learning pathways in student behavior as **learning patterns**, which indicates that student learning paths are not random but follow discernible patterns. Figure 1 shows an example. Following these learning patterns in high-quality data generative offers potential to simplify the complexity of model-centric paradigms and enhance interpretability. However, this is non-trivial, which necessitates addressing two challenges: (1) How should we systematically extract and effectively utilize these high-quality data sources to generate accurate and informative exercise sequence data for enhancing model training, and through what mechanisms can we reliably integrate such data into the training dataset. (2) How to design response labels for generated exercise sequence data, which means the knowledge states of students regarding the generation data is unknown, therefore, new exercise sequences cannot be accurately labeled for student response.

To address these challenges, we propose a learning patterns-guided data generation framework for regenerating reliable student interaction data. Our approach consists of two key components: (1) a pair-wise mapping strategy to mine learning patterns from the original data, providing a common representation for high-quality generative,then an encoder-decoder architecture to capture and regenerate student learning patterns; and (2) a diffusion-guided pseudo-label generator to enhance the diagnostic value of generated sequence data by evolving latent knowledge states. Extensive experiments demonstrate the effectiveness of our framework.

2 Related Work

2.1 Knowledge Tracing

Knowledge tracing, a core component of intelligent tutoring systems originating in 1994, forecasts student performance through exercise interactions. Main approaches include probabilistic statistical methods and deep learning architectures [17, 21, 35]. Early Bayesian Knowledge Tracing (BKT [1]) employed hidden Markov models (HMMs) to model students' knowledge states [7, 34, 42]. Deep learning models, on the other hand, typically leverage sequential models. For instance, DKT([27]) introduced LSTM to model student knowledge states but often suffered from overfitting due to the exercise of interaction data sparsity.

In order to solve the interaction data sparsity, early studies revolved around model-centric paradigm. Existing studies have focused on approaches as followed: The knowledge tracing model that integrates attention alleviates the sparse interaction to a certain extent. For example, SAKT([25]) and AKT etc. [6, 19, 23, 25, 41] using attention architecture. Secondly, representation enhancement techniques have been explored, such as GIKT([37]), which aggregates exercise and concept information through graph neural networks, PEBG([18]), which uses pretraining to extract latent relationships between exercises and concepts, and CL4KT([13]), which designs multiple enhancement strategies combined with contrastive learning to strengthen student interactions and exercise representations. DKTMR([2]) combines graph learning and attention mechanisms. It uses DGGAN and GraphGAN for relation graphs, updates states with GRU, and fuses relations via attention, improving prediction and interpretability. QRCL([46]) utilizes a dual-contrast approach that combines native interactions, SVD-enhanced views,

and inter-exercise collaborative relationship views, all underpinned by graph neural networks to effectively generate exercise-answer representations.

While existing methods [24, 36, 40, 43] partially mitigate model-centric data sparsity, low-quality interaction data persists as a key constraint in knowledge tracing. We propose a data-centric framework that extracts learning behaviors, trains pattern-capturing models, and generates enhanced interaction sequences to alleviate sparsity and boost model performance.

2.2 Data-Centric in Recommendation System

With the rapid development of artificial intelligence technology, the key role played by data in determining model performance has become increasingly prominent. In the field of recommendation systems, the data-centric development paradigm can solve the problem of data sparsity and improve model performance. In this data-centric development paradigm, data generation plays a key role. The key is to generate new data with the help of original data to enhance the performance of existing sequential recommendation models. Among them are data generated based on models. For example, the UPC-SDG([15]) generates interaction data that meets user privacy preferences, solving the privacy issues that may be caused by the collection of user behavior data. The DR4SR([39]) model uses Transformer to generate the optimal sequence recommendation dataset, significantly improving the performance of the sequence recommendation task. The ∞-AE([28]) model uses a neural tangent kernel approximation to model an infinite width autoencoder and generates users through a sampling-based reconstruction method. Diff4Rec([33]) pre-trains a diffusion model on the recommendation data and uses the generated predictions to generate diverse enhancements for sparse user-item interactions. DiffuASR([16]) is a diffusion-based pseudo-sequence generation framework in which the extended dataset can be directly used to train sequential recommendation models. Another way to generate data samples is data distillation. A data distillation framework called DISTILL-CF is proposed ([28]) to create small data for collaborative filtering data and generate a compact dataset by using a two-layer optimization objective and a differentiable Gumbel sampling technique. FARZI([29]) aims to summarize large-scale autoregressive datasets by generating a small number of generative sequences to maintain or improve model performance. GCM([31]) converts discrete categorical data into a continuous data format and adopts a single-step gradient matching strategy to efficiently distill a small generative dataset that can be used to train a CTR model.

While these data generation techniques in sequential recommendation provide valuable insights, knowledge tracing poses unique challenges that require specialized solutions, as we described in the Section 1.

3 Problem Definition

In this section, we formally define the task of knowledge tracing. Let the original training set be denoted as $\mathcal{D}$, including that $\mathcal{S} = \{s_1, s_2, \ldots, s_U\}$ be the set of U students, $\mathcal{E} = \{e_1, e_2, \ldots, e_M\}$ be the set of M exercises. We represent the exercise-solving sequence for each student throughout the learning process as $\mathcal{R} = (e_1, r_1), (e_2, r_2), \ldots, (e_T, r_T)$, where the (e_i, r_i) is the i-th interaction behavior, and $r_i \in \{0, 1\}$ represents the response to exercise e_i, $r_i = 1$ means the student answered correctly, $r_i = 0$ means the student answered incorrectly.

Our approach focuses on constructing high-quality training data $\mathcal{D}'$ to enhance knowledge tracing performance. Hence, we strive to reformulate the specific task into a data generation exercise, which is formally defined as follows:

Definition 1.(Data Generation). Given the sequence $\mathcal{R} \in \mathcal{D}$, data generation aims to learn a sequence regenerator f_r and a pseudo-label generator f_g, which constitute high-quality training data $\mathcal{D}_{\text{hq}}$ guided by the learning patterns. By merging $\mathcal{D}$ and $\mathcal{D}_{\text{hq}}$, we can obtain the new training data $\mathcal{D}'$.

Definition 2.(Knowledge Tracing). Given student's historical interaction sequence $\mathcal{R} \in \mathcal{D}'$, our goal is to trace the student's knowledge states and predict student answer performance at next time.

4 Methodology

In this section, we first present a general overview of our proposed LPDG framework. As shown in Figure 2, our LPDG framework comprises two key components: the learning patterns-guided exercise sequence regenerator and the learning patterns-guided pseudo-label generator.

4.1 Learning Patterns-Guided Exercise Sequence Regenerator

Understanding and capturing the changes in students' learning patterns is crucial in traditional knowledge tracing task [5, 45]. Similarly, to effectively extract students' latent learning patterns, we carefully design three submodules: the learning patterns pair-wise mapping, the contextual pre-training regenerator, and the exploratory inference strategy, which are detailed as follows.

4.1.1 Learning Patterns Pair-Wise Mapping. Inspired by [32, 38], by establishing a precise rule-based mapping mechanism, we can systematically correlate students' learning behaviors with changes in their knowledge states, thereby providing a more detailed and comprehensive analysis of their learning trajectories.

Specifically, we employ a sliding window strategy to extract learning patterns. By sliding a window of size π, we generate subsequences from the sequences within the window. Then, we count the frequency of response patterns to various exercise subsequences from all the original response sequences. Any subsequence that appears more than the predefined threshold ϵ is selected and utilized to construct data pairs. These data pairs ultimately form a new paired set $\mathcal{D}_{\text{pair}}$. Figure 3 illustrates an example of this process.

After obtaining the paired set $\mathcal{D}_{\text{pair}}$, the training data can be constructed into sequence pairs, denoted $(\mathcal{R}, \mathcal{R}_i^{\text{lp}}), i \in \{1, 2, \ldots, N\}$, where $\mathcal{R}_i^{\text{lp}}$ represents the learning patterns mined through the precise rule-based mapping mechanism and N represents the number of learning patterns.

4.1.2 Contextual Pre-training Regenerator. After obtaining the constructed original sequences and learning patterns pairs $(\mathcal{R}, \mathcal{R}_i^{\text{lp}})$, we leverage the powerful generative capability and contextual awareness of the transformer to construct a pre-training task for subsequent sequence regeneration. Specifically, we initially transform the

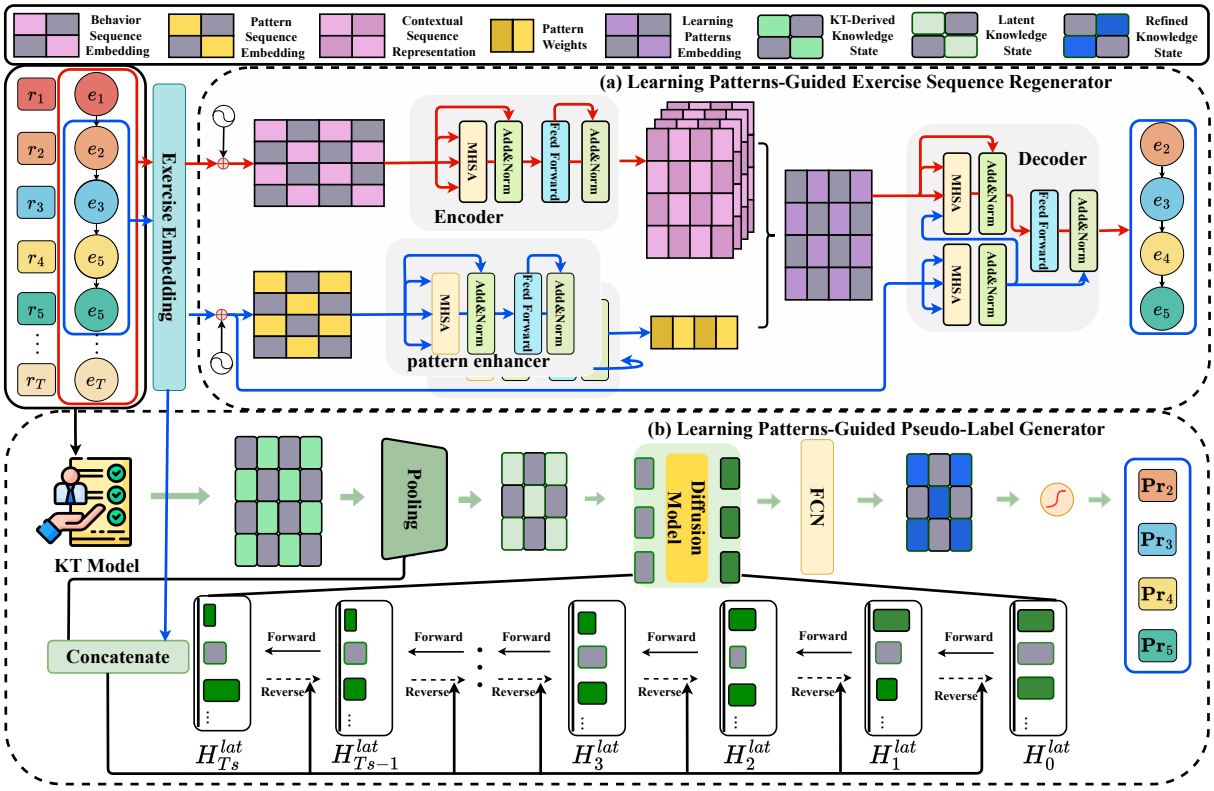


Figure 2: The overview architecture of our proposed LPDG framework.

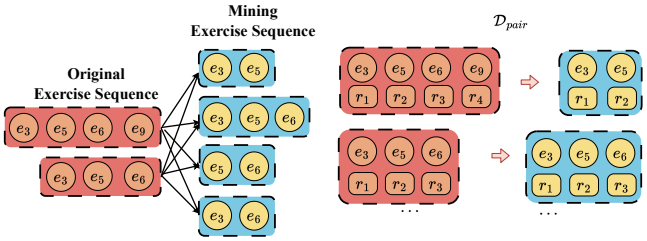


Figure 3: The illustration shows the schematic diagram of the subsequences generated with a sliding window of 3. Given the sequence (e_3, e_5, e_6, e_9) and the sequence (e_3, e_5, e_6), the following subsequences can be extracted: $(e_3, e_5)(e_3, e_6)(e_5, e_6)(e_3, e_6, e_9)$

exercise sequence into behavior sequence embedding $\mathbf{E}^b \in \mathbb{R}^{T\times d}$ through embedding methods, where T represents the sequence length and d is the embedding size. Then, we feed the behavior sequence embedding $\mathbf{E}^b$ into the encoder of the transformer:

$$\mathbf{E}^c = \text{Encoder-Layer}(\mathbf{E}^b), \tag{1}$$

where $\mathbf{E}^c \in \mathbb{R}^{T\times d}$ represents the contextual sequence representation and Encoder-Layer$(\cdot)$ represents one layer of transformer encoder.

After computing the student's contextual sequence representation $\mathbf{E}^c$ from the original exercise sequence, we project it into K different pattern spaces, each corresponding to a specific learning pattern. This mapping approach effectively captures the correlation between contextual information and learning patterns. Subsequently, we obtain the contextual sequence representation $\mathbf{E}^c_1, \mathbf{E}^c_2, \ldots, \mathbf{E}^c_K$ in different spaces.

However, there exists a one-to-many relationship between $\mathcal{R}$ and $\mathcal{R}^{\text{lp}}$ in the paired set $\mathcal{D}_{\text{pair}}$, we employ a tailor-made pattern enhancer, designed to strengthen the expressiveness of learning patterns and further enhance the model's understanding of students' response behaviors. Specifically, we can obtain the pattern weights $\boldsymbol{\theta} \in \mathcal{R}^K$ by:

$$\boldsymbol{\theta} = \text{Softmax}(\text{MLP}(\text{Pattern-Enhancer}(\mathbf{E}^p))), \tag{2}$$

where $\mathbf{E}^p \in \mathbb{R}^{T^{\text{lp}}\times d}$ denotes the initial pattern sequence embeddings we obtained, T^{lp} is the maximum sequence length of all learning patterns, and Pattern-Enhancer$(\cdot)$ denotes two layer of transformer encoder. MLP$(\cdot)$ denotes the multilayer perceptron and Softmax$(\cdot)$ denotes the softmax function. Next, we can obtain the learning patterns embedding $\mathbf{E}^{\text{lp}}$ by:

$$\mathbf{E}^{\text{lp}} = \sum_{i=1}^{K} \theta_i \mathbf{E}^c_i. \tag{3}$$

To effectively guide the parameter learning of the regenerator, we incorporate the decoder component of the transformer, with the primary objective of reconstructing learning patterns. Through this meticulously designed pre-training task, we can leverage the

exploratory inference strategy to generate new high-quality data. Thus, we can obtain the loss function for training the exercise sequence generator as follows:

$$L_{\text{lp}} = -\sum_{(\mathcal{R},\mathcal{R}^{\text{lp}})}^{|\mathcal{D}_{\text{pair}}|}\sum_{l=1}^{L}\log P(e_{il}|\mathbf{E}^c,\hat{\mathcal{R}}^{\text{lp}}_{<l}), \tag{4}$$

where l is the position in the target pattern, L is the number of learning patterns, $\hat{\mathcal{R}}^{\text{lp}}_{<l}$ represents the learning pattern generated before l, and $P(e_{il}|\cdot)$ denotes the conditional probability of the target exercise e_{il}.

4.1.3 Exploratory Inference Strategy. To thoroughly explore students' latent learning preferences and knowledge mastery, this strategy drives the regenerator to infer diverse and unobserved exercise sequences. Specifically, the strategy leverages the encoder and pattern enhancer to obtain alternative contextual sequence representation $\mathbf{E}^{\text{lp}}_{\text{infer}}$ distinct from the original input $\mathbf{E}^b$. A masking mechanism is introduced during the decoding phase to encourage the model to infer diverse exercise paths while maintaining consistency with the extracted patterns, fully embodying the core concept of exploration. Ultimately, we can regenerate entirely new high-quality exercise sequences $\mathcal{R}^{\text{hq}}$; however, these sequences are not accompanied by labels, necessitating the supplementation of annotated information through pseudo-label construction.

4.2 Learning Patterns-Guided Pseudo-Label Generator

In the previous section, we have regenerated entirely new and high-quality sequences $\mathcal{R}^{\text{hq}}$. To annotate these sequences with labels that reflect the true responses of students, we have meticulously designed a learning patterns-guided pseudo-label generator to simulate the evolution of knowledge states across these sequences. The generator consists of two submodules: the diffusion-based probability estimation and the pseudo-label mapping strategy. Detailed explanations are provided below.

4.2.1 Diffusion-Based Probability Estimation. To ensure the quality of the labels, we employ a pre-trained KT model to extract knowledge states, guiding the high-quality generation of pseudo-labels. Specifically, we first input the student's original sequence $\mathcal{R}$ into the pre-trained KT model, and then it outputs the KT-derived knowledge state $\mathbf{H}^{\text{pre}} \in \mathbb{R}^{T\times d}$. Then, it is fed into the pooling layer, where the pooling is performed along the time dimension to capture the features of students' knowledge state transitions, denoted as latent knowledge state $\mathbf{H}^{\text{lat}} \in \mathbb{R}^{T^{\text{lp}}\times d}$ where T^{lp} is the maximum sequence length of all learning patterns.

Forward Process. We employ the forward process of the diffusion model([8]) to incrementally add noise to the latent knowledge state $\mathbf{H}^{\text{lat}}$. By doing so, the data distribution in the latent space gradually transitions toward a standard Gaussian distribution, laying the foundation for the subsequent reverse denoising process, thereby generating more robust and reliable pseudo-labels. Let $\mathbf{H}^{\text{lat}}_0 \sim q(\mathbf{H}^{\text{lat}}_0)$ denotes the original data samples. This process can be expressed as follows:

$$q(\mathbf{H}^{\text{lat}}_{1:T_s}|\mathbf{H}^{\text{lat}}_0) = \prod_{t=1}^{T_s}\mathcal{N}(\mathbf{H}^{\text{lat}}_t;\sqrt{\bar{\alpha}_t}\mathbf{H}^{\text{lat}}_0,(1-\bar{\alpha}_t)\mathbf{I}), \tag{5}$$

where $\alpha_t = 1-\beta_t$, $\bar{\alpha}_t = \prod_{s=1}^{t}\alpha_s$, β_t denotes the variance schedule and T_s denotes the diffusion step.

Reverse Process. The reverse process aims to recover the original data sample $\mathbf{H}^{\text{lat}}_0$ from the completely perturbed $\mathbf{H}^{\text{lat}}_{T_s} \sim \mathcal{N}(0,\mathbf{I})$. The reverse process can be formulated using the Markov chain as follows:

$$\begin{aligned} p_\phi(\mathbf{H}^{\text{lat}}_{t-1}|\mathbf{H}^{\text{lat}}_t) &= \mathcal{N}(\mathbf{H}^{\text{lat}}_{t-1};\mu_\phi(\mathbf{H}^{\text{lat}}_t,\mathbf{C},t),\textstyle\sum_\phi(t)),\\ \mu_\phi(\mathbf{H}^{\text{lat}}_t,\mathbf{C},t) &= \frac{1}{\sqrt{\alpha_t}}(\mathbf{H}^{\text{lat}}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\phi(\mathbf{H}^{\text{lat}}_t,\mathbf{C},t)),\\ \mathbf{C} &= (\mathbf{H}^{\text{lat}}\oplus\mathbf{E}^p)\mathbf{W}_1+\mathbf{b}_1, \end{aligned} \tag{6}$$

where $\mu_\phi(\cdot)$ and $\sum_\phi(t)$ are the mean and variance that are parameterized by ϕ. $\epsilon_\phi(\cdot)$ denotes a parameterized linear layer. $\oplus$ denotes the concatenation operation, $\mathbf{W}_1 \in \mathbb{R}^{2d\times d}$ and $\mathbf{b}_1 \in \mathbb{R}^d$ are the trainable parameters.

The generated refined knowledge state $\mathbf{H}^{\text{re}}$ can be computed by the following:

$$\mathbf{H}^{\text{re}} = \mathbf{H}^{\text{lat}}_0\mathbf{W}_2+\mathbf{b}_2, \tag{7}$$

$$\mathbf{Pr} = \sigma(\mathbf{H}^{\text{re}}\mathbf{W}_3+\mathbf{b}_3), \tag{8}$$

where $\mathbf{W}_2,\mathbf{W}_3 \in \mathbb{R}^{d\times d}$, $\mathbf{b}_2,\mathbf{b}_3 \in \mathbb{R}^d$ are the trainable parameters. $\mathbf{Pr}$ are the probability of answering correctly.σ is the sigmoid activation function.

The goal of the diffusion model ([8]) is to bring the posterior distribution of the forward process closer to the prior distribution from the reverse process. Therefore, the objective function can be written in the form of a KL divergence. To simplify, it can also be formalized as:

$$L_{\text{Diffusion}} = \mathbb{E}_{t,\mathbf{H}^{\text{lat}}_0,\epsilon}\left[\left\|\epsilon-\epsilon_\phi(\sqrt{\bar{\alpha}_t}\mathbf{H}^{\text{lat}}_0+\sqrt{1-\bar{\alpha}_t}\epsilon,t)\right\|^2\right] \tag{9}$$

Meanwhile, we employ a sequence task to predict the correctness of responses on the learning pattern sequences, optimizing the parameters of the learning patterns-guided pseudo-label generator, which can be represented as follows:

$$L_{\text{lp-pred}} = -\sum_{i=1}^{T^{\text{lp}}}(r_i\log(\mathbf{Pr}_i))+(1-r_i)\log(1-\mathbf{Pr}_i) \tag{10}$$

Finally, the total loss L_{gene} for this section is balanced by γ, which adjusts the weights between the $L_{\text{Diffusion}}$ and $L_{\text{lp-pred}}$:

$$L_{\text{gene}} = L_{\text{Diffusion}} + \gamma L_{\text{lp-pred}}. \tag{11}$$

4.2.2 Pseudo-Label Mapping Strategy. Through the exploratory inference strategy, we can obtain new high-quality exercise sequences $\mathcal{R}^{\text{hq}}$. Subsequently, we can infer the missing label information of $\mathcal{R}^{\text{hq}}$ (that is, the probability of $\mathbf{Pr}$ predicted by diffusion-based probability estimation) using the well-trained learning pattern-guided pseudo-label generator.

Errors in data-driven pseudo-label mapping can cause bias. Based on the monotonicity assumption that practice improves ability. Therefore, we designed a dynamic method for pseudo-label mapping, as shown below: Specifically, after obtaining the value of

the student's estimated correct answer, we first set a threshold ε ($0 < \varepsilon < 1$) for the answer probability $\mathbf{pr}_i$ at each moment to map the pseudo-label according to the following rules:

$$\hat{r}_i = \begin{cases} 0, & \mathbf{pr}_i \leq \varepsilon \\ 1, & \mathbf{pr}_i > \varepsilon. \end{cases} \tag{12}$$

Each time $\hat{r}_i = 0$ is mapped, the value of ε is reduced by η until ε is restored to its original value when $\hat{r}_i = 1$;

After generating the new exercise sequences $\mathcal{R}^{\text{hq}}$ and their corresponding labels, We integrate it with the original data $\mathcal{D}$ to form a new training dataset $\mathcal{D}'$. Subsequently, we apply it to downstream KT tasks to evaluate the quality of the generated data and validate the effectiveness of the model. Given a exercise interaction sequence $s_i \in \mathcal{D}'$, the loss function of the downstream task is as follows:

$$L_{\text{KT}} = -\sum_{t=1}^{T} (r_t^{s_i} \log \hat{r}_t^{s_i} + (1 - r_t^{s_i}) \log(1 - \hat{r}_t^{s_i})) + \lambda \left\| \Theta \right\|_2^2, \tag{13}$$

where T is the length of s_i, λ represents the hyperparameter of L_2 regularization strength and Θ is the set of all model parameters.

5 Experiment

Table 1: Statistics of three real-world datasets in this paper.

Dataset	Student	Exercise	Interaction	Sparsity
Assistment09	3.6k	17.7k	282.0k	0.9955
Assistment12	25.3k	50.9k	2621.3k	0.9979
Assistment17	1.7k	3.2k	942.8k	0.8266

In this section, we conducted experiments on three real-world datasets to evaluate our proposed LPDG framework. Specifically, we aim to address the following research questions:

- **RQ1**: What is the performance of the LPDG framework on datasets characterized by sparse interaction data?
- **RQ2**: Does our data generation method outperform other generation methods in terms of data quality?
- **RQ3**: Can experimental results validate the effectiveness of the various components within LPDG?
- **RQ4**: Compared to the original dataset, does the generated dataset better align with the evolution patterns of students' knowledge states?

5.1 Experimental Setting

5.1.1 Dataset Description. We conducted extensive experiments on three real-world datasets, including Assistment09, Assistment12 and Assistment17. Table 1 shows the statistics of the datasets. The calculation formula of Sparsity in table 1 is:

$$Sparsity = 1 - \frac{Interaction}{Student \cdot Exercise} \tag{14}$$

80% of the learning sequences from each complete dataset were divided into a training dataset (70%) and a validation dataset (10%), with the remaining 20% reserved for the test dataset. In the experiment, we mainly used Assistment09, Assistment12, and Assistment17. The details of the dataset are shown in the table1. In order to optimize the computational efficiency, we set the original student sequence interaction to set the maximum length of the student's learning sequence to 50, and truncate the learning sequence exceeding this length into multiple subsequences. To ensure rationality, we screened out sequences with a length less than 5. The following is a description of the dataset:

- **Assistment09** [3]: Originating from the 2009 Assistment project, this dataset is primarily used to analyze students' performance and behavior in mathematics learning, supporting educational technology research and achievement prediction.
- **Assistment12** [4]: Derived from the expanded Assistment system in 2012, this dataset aims to provide data support for personalized learning systems and learning process modeling.
- **Assistment17** [26]: Sourced from the updated Assistment platform in 2017, this dataset is primarily used for educational technology research, machine learning applications, and the detailed modeling of learning processes.

5.1.2 Baseline Model and Evaluation Metrics. To evaluate the effectiveness of the proposed LPDG framework, we selected several state-of-the-art baselines as comparison methods, including SAKT [25], AKT [6], CL4KT [13], simpleKT [19], sparseKT [10], FoLiBi-AKT [11] and stableKT [14].

- **SAKT [25]**: SAKT model uses self-attention to efficiently predict student performance by identifying relevant past interactions.
- **CL4KT [13]**: CL4KT model employs contrastive learning to improve knowledge tracing by distinguishing semantically similar and dissimilar student learning histories. It uses transformer-based encoders with educationally motivated data augmentations.
- **AKT [6]**: AKT leverages context-aware embeddings and a modified Transformer architecture with adaptive attention weights calculated through distance-aware exponential decay.
- **FoLiBi-AKT [11]**: FoLiBi decouples exercise relevance and forgetting behavior by introducing a linear bias, which can effectively reflect the impact of forgetting behavior on the learner's knowledge level. We use its AKT variant for experiments, and named FoLiBi-AKT.
- **simpleKT [19]**: simpleKT captures individual differences in exercises covering the same knowledge points by explicitly modeling exercise-specific difficulty differences, and uses a common dot-product attention function to extract temporal information in students' learning interactions.
- **sparseKT [10]**: sparseKT enhances knowledge tracing by incorporating k-sparse attention mechanisms to selectively focus on the most influential historical interactions.
- **stableKT [14]**: stableKT is an attention-based knowledge tracing model that enhances generalization to long sequences by incorporating linear biases and employs a multi-head aggregation module to capture complex relationships.

Table 2: The performance of all baselines was evaluated on three real-world datasets using both the original datasets and the datasets generated by the LPDG framework. The best results are shown in bold.

Method		Assistment09		Assistment12		Assitment17	
		ACC	AUC	ACC	AUC	ACC	AUC
CL4KT	original	$0.7348_{\pm 0.0050}$	$0.7608_{\pm 0.0050}$	$0.7295_{\pm 0.0010}$	$0.7212_{\pm 0.0010}$	$0.6961_{\pm 0.0010}$	$0.7306_{\pm 0.0010}$
SAKT	original	$\mathbf{0.6981_{\pm 0.0030}}$	$0.6796_{\pm 0.0030}$	$0.7183_{\pm 0.0010}$	$0.6632_{\pm 0.0004}$	$0.6713_{\pm 0.0010}$	$0.6692_{\pm 0.0005}$
	LPDG	$0.6968_{\pm 0.0010}$	$\mathbf{0.6831_{\pm 0.0010}}$	$\mathbf{0.7192_{\pm 0.0010}}$	$\mathbf{0.6645_{\pm 0.0005}}$	$\mathbf{0.6833_{\pm 0.0010}}$	$\mathbf{0.7196_{\pm 0.0008}}$
AKT	original	$\mathbf{0.7200_{\pm 0.0040}}$	$0.7466_{\pm 0.0030}$	$0.7296_{\pm 0.0020}$	$0.7210_{\pm 0.0006}$	$0.6939_{\pm 0.0010}$	$0.7235_{\pm 0.0009}$
	LPDG	$0.7170_{\pm 0.0050}$	$\mathbf{0.7490_{\pm 0.0020}}$	$\mathbf{0.7319_{\pm 0.0020}}$	$\mathbf{0.7306_{\pm 0.0008}}$	$\mathbf{0.6942_{\pm 0.0010}}$	$\mathbf{0.7279_{\pm 0.0010}}$
FoLiBi-AKT	original	$0.7146_{\pm 0.0040}$	$0.7448_{\pm 0.0010}$	$0.7308_{\pm 0.0020}$	$0.7212_{\pm 0.0010}$	$0.6951_{\pm 0.0020}$	$0.7327_{\pm 0.0010}$
	LPDG	$\mathbf{0.7210_{\pm 0.0040}}$	$\mathbf{0.7503_{\pm 0.0010}}$	$\mathbf{0.7352_{\pm 0.0004}}$	$\mathbf{0.7312_{\pm 0.0005}}$	$\mathbf{0.6966_{\pm 0.0010}}$	$\mathbf{0.7343_{\pm 0.0010}}$
simpleKT	original	$0.7080_{\pm 0.0040}$	$0.7269_{\pm 0.0010}$	$0.7247_{\pm 0.0010}$	$0.7008_{\pm 0.0010}$	$0.7099_{\pm 0.0010}$	$0.7585_{\pm 0.0020}$
	LPDG	$\mathbf{0.7254_{\pm 0.0010}}$	$\mathbf{0.7471_{\pm 0.0030}}$	$\mathbf{0.7410_{\pm 0.0003}}$	$\mathbf{0.7393_{\pm 0.0010}}$	$\mathbf{0.7120_{\pm 0.0010}}$	$\mathbf{0.7617_{\pm 0.0010}}$
sparseKT	original	$0.7040_{\pm 0.0050}$	$0.7178_{\pm 0.0050}$	$0.7252_{\pm 0.0007}$	$0.7010_{\pm 0.0001}$	$0.7103_{\pm 0.0010}$	$0.7600_{\pm 0.0010}$
	LPDG	$\mathbf{0.7235_{\pm 0.0020}}$	$\mathbf{0.7465_{\pm 0.0020}}$	$\mathbf{0.7401_{\pm 0.0010}}$	$\mathbf{0.7392_{\pm 0.0002}}$	$\mathbf{0.7134_{\pm 0.0010}}$	$\mathbf{0.7633_{\pm 0.0010}}$
stableKT	original	$0.7149_{\pm 0.0040}$	$0.7355_{\pm 0.0020}$	$0.7249_{\pm 0.0010}$	$0.7009_{\pm 0.0001}$	$0.7133_{\pm 0.0010}$	$0.7639_{\pm 0.0010}$
	LPDG	$\mathbf{0.7239_{\pm 0.0009}}$	$\mathbf{0.7462_{\pm 0.0010}}$	$\mathbf{0.7412_{\pm 0.0010}}$	$\mathbf{0.7396_{\pm 0.0004}}$	$\mathbf{0.7141_{\pm 0.0003}}$	$\mathbf{0.7650_{\pm 0.0008}}$

Following the previous works, we employed two widely-used metrics to evaluate the effectiveness of our framework, including **Prediction Accuracy (ACC)** and **Area Under an ROC Curve (AUC)**.

5.1.3 Implementation Details. We first generalized the data-fair dataset and then applied it to each model fairly. The knowledge tracing model's hyper-parameters of comparison approaches are tuned on the validation set according the original paper. Experimental results for all models are obtained by performing standard runing 5 times. And take the mean and variance as the display. When we train the diffusion-based probability estimation module, we set the balance parameter γ to 0.5. we searched the ε from {0.5, probability average value} and the η from {0.0001,0.003} for pseudo-label mapping of the correct answer probability of the new sequence students at each moment. When training Diffusion-Based Probability Estimation and generating potential knowledge states, we set the number of forward and reverse diffusion steps $T_s = 1000$. To accelerate the generation speed, we used dpm-slover++([20]). We introduce the data generated by our framework into the knowledge tracing models for training. Since the sparsity and sequence number of each dataset is different, we set π to 20 and ϵ to 2 when extracting learning patterns on Assistment09, and set π to 10 and ϵ to 10 when extracting learning patterns on Assistment12 and Assistment17 datasets. In order to ensure the effective learning of our framework, we only extract learning pattern sequences with a length of more than 10 for Section 4.1.1. When training all knowledge tracing models, we searched the λ in $\{1e-5, 2e-5, 3e-5\}$. We initialized all the parameters in the networks with Xavier initialization and used the Adam optimizer during the training process. All experiments are conducted on an NVIDIA RTX4090 GPU. Our code can be found at https://github.com/BIMK/Intelligent-Education/tree/main/LPDG.

5.2 Overall Performance (RQ1)

To validate the effectiveness of our proposed LPDG framework on datasets with sparse interactions, we applied the LPDG framework to the Assistment09, Assistment12 and Assistment17, datasets. Specifically, we trained KT models using both the generated training dataset by LPDG and the original training dataset, and evaluated their performance on real interaction data.

The experimental results are presented in the table 2. The results demonstrate the following key observations: Models trained with the training dataset generated by the LPDG framework outperform those trained with the original training datasets. Specifically, we observe the following phenomena: The performance improvement is most pronounced on the Assistment09 and Assistment12 datasets. This improvement may stem that the LPDG-generated training dataset yield enhanced interaction diversity, which facilitates superior generalization of models to unseen interactions. In comparison, the Assistment17 dataset contains fewer exercises and exhibits lower sparsity in student-exercise interactions, resulting in less pronounced performance improvements compared to the first two datasets. The performance gains of simpleKT, sparseKT, and stableKT are markedly superior compared to other models. This distinction arises from their architecture, which prioritizes domain-specific pattern extraction and inherently aligns with generative data characteristics. The simpleKT explicitly models exercise-specific variations, enabling it to better capture the variations in learning patterns in the generated data and performs better than the original dataset. The sparseKT selects key interactions through a sparse attention mechanism, reducing noise interference, and the clear diversity in the generative data enhances its performance; The stableKT optimizes length generalization and hierarchical dependency modeling via linear biases and other methods. The extended

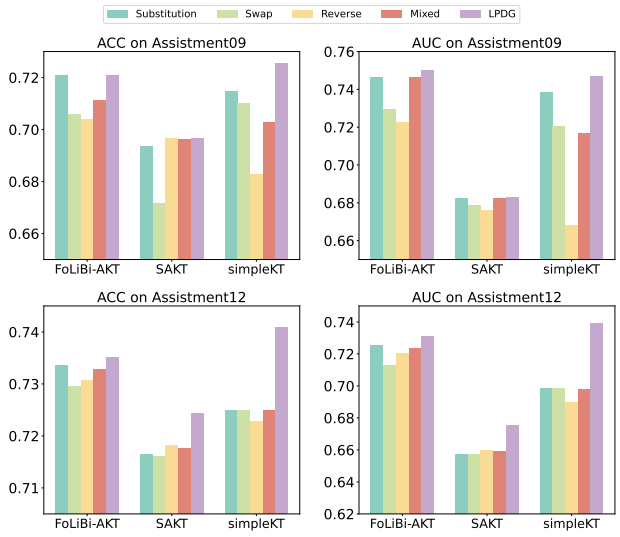


Figure 4: A comparison of the performance of training knowledge tracing models using generated training data from other methods and that from LPDG framework generated training data.

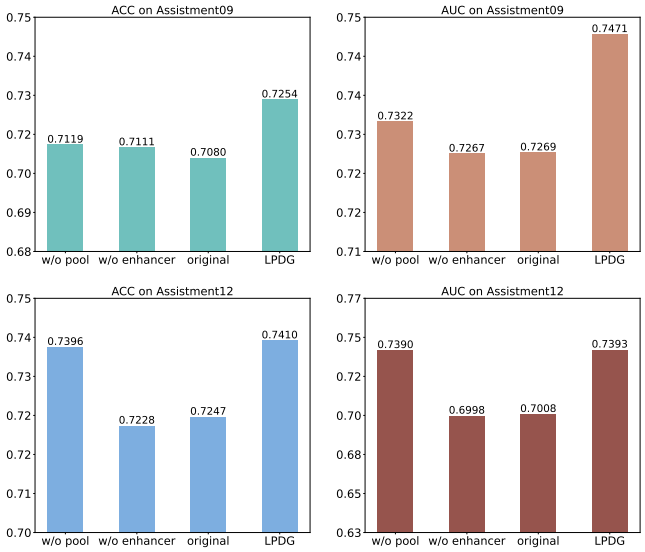


Figure 5: Performance of ablation studies conducted on the Assistment09 and Assistment12.

sequence length and exercise hierarchies embedded in generative data further refine its capability to capture long-range knowledge dependencies. SAKT shows low sensitivity to data enhancements. FoLiBi-AKT and AKT, heavily context-dependent, adapt poorly to generated data.

5.3 Quality Verification (RQ2)

To evaluate the interaction sequences generated by LPDG, we compared its performance with other sequence generation methods on the sparse datasets Assistment09 and Assistment12. **Substitution**: Replace the exercises in the sequence with an exercise of similar accuracy rate with a probability of 0.3 to generate a new sequence that combines the original training dataset. **Reverse**: Reversing student response correctness labels with 0.4 probability to generate a new sequence merged the original training dataset. **Swap**: Exchange adjacent motion-responses with a probability of 0.5 to generate a new sequence merged the original training dataset. **Mixed**: Combining the above three strategies with weighting ratios of 0.3, 0.3, and 0.4, respectively, generate a new sequence merged the original training dataset. we use the simpleKT, FoLiBi-AKT, and SAKT as base KT models.

Experimental results (illustrated in the figure 4) reveal two key observations: **Substitution** achieved optimal performance by diversifying exercises through strategies, enhancing interaction diversity for better generalization. **LPDG** outperforms other strategies through explicit modeling learning patterns, and the generated data performs better on most models. Unlike rule-based methods, it preserves pedagogical coherence via students' learning pattern simulation of authentic learning behaviors. This demonstrates LPDG's dual enhancement of interaction diversity and educational validity through learning-pattern sequence generation.

5.4 Ablation Study (RQ3)

To rigorously validate the functional contributions of key modules in our framework, we conducted module ablation experiments using simpleKT as the KT model on the Assistment09 and Assistment12 datasets. The experimental configurations were defined as follows: **LPDG**: KT models trained on datasets generated by **LPDG**; **Original**: KT Models trained on the original training dataset; **w/o pool**: Removal of the pooling module in the Diffusion-Based Probability Estimation; **w/o enhancer**: Disabling the tailor-made pattern enhancer during exploratory inference.

The experimental results (illustrated in the figure 5) demonstrate the following conclusions: Removing any module leads the KT model to performance degradation, confirming the critical roles of all components. **w/o pool** introduces partial inaccuracies in localized knowledge states within the generated exercise sequences, resulting in erroneous student feedback. However, due to the continuity of the learning process, these localized inaccuracies do not significantly disrupt the global evolution of knowledge states, leading to only marginal performance degradation compared to the original training dataset. **w/o enhancer** prevents the sequence regenerator capturing the diversity of student learning patterns. This limitation reduces interactivity and heterogeneity in generated sequences, causing significant performance decline.

5.5 Case Study (RQ4)

This figure 6 visualizes the knowledge states generated by the simpleKT in the first 30 time steps of a interaction sequence, where the simpleKT are trained based on different training datasets. Left (Original Dataset) vs. Right (LPDG-Generated): simpleKT knowledge states comparison in figure 6. The horizontal axis and vertical axis represent the time steps and the concept vector respectively. The knowledge states generated by the model trained based on LPDG generated training dataset, showing stronger temporal coherence than the model trained with original dataset. And concept 1 and

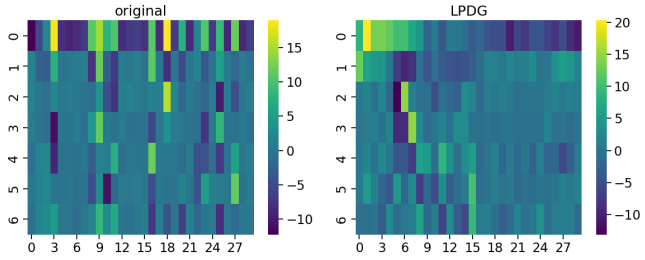


Figure 6: We randomly selected the first 30 records in an interaction sequence, used the knowledge tracing model trained with the original training dataset and our generation training dataset to output their knowledge states matrix, and used PCA to reduce the knowledge states at each moment to the dimension of concept.

2, etc. at specific time steps show a clearer and more concentrated trend. By increasing the diversity and coherence of sequence data, generated data not only simulates more realistic student learning behaviors, but also improves the model's ability to capture dynamic changes, and expands the coverage of knowledge states.

5.6 Parameter Sensitivity Analysis

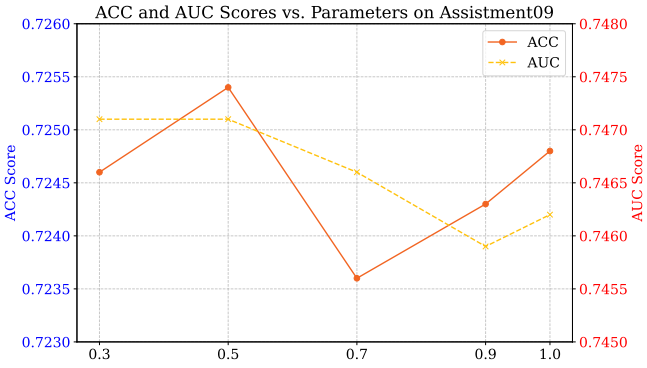


Figure 7: Sensitivity analysis of γ on Assistment09

In this experiment, we use simpleKT [19] as the base model to test the performance of generated datasets in the Assistment09 dataset with different parameters γ. From the figure 7, we can draw the following conclusions: when γ=0.7 or γ=0.9 , the model completely favors the student feedback loss, resulting in the neglect of knowledge state generation; and when γ=0.5, the generated dataset performs best, the weights of the two are balanced, and the model achieves the best generation performance between the two tasks.

5.7 Time Efficiency Analysis

The time mainly spent by the knowledge tracing model using the LPDG framework consists of two parts: data generation and model training. In the data generation stage, new exercise sequences are generated through the reasoning of the two main components. In terms of generation efficiency, on the NVIDIA RTX 4090 with the batchsize set to 125, the generation time for each sequence is approximately 0.5 seconds.

Table 3: We mainly use simpleKT to analyze the performance gains and time complexity brought by using the original training data and the training data generated by LPDG.

dataset	method	runtime s/epoch	ACC Improve	AUC Improve
Assistment09	original	7.2	0.0174	0.0202
	LPDG	11.72		
Assistment12	original	36.64	0.0163	0.0385
	LPDG	40.15		
Assistment17	original	26.38	0.0021	0.0032
	LPDG	28.68		

The tabel 3 shows the performance gains and time complexity brought by simpleKT mainly using the original training data and the training data generated by LPDG. From the tabel 3, we can draw the following viewpoints:

On the Assistment09 and Assistment12 datasets, the model achieved significant performance improvements after using the training data synthesized by LPDG. This is because the generated data effectively alleviated the extreme sparsity of the original data, enabling the model to better capture learning patterns and only increasing the time cost of moderate training. On the Assistment17 dataset, the original data shows a lower sparsity, resulting in a smaller marginal benefit of the generated data. Although, it shows increase in time cost, LPDG achieves a reasonable balance between generation time and performance gain in sparse scenarios. It is worth noting that the data generation process is decoupled from the training process of the knowledge tracing model below and can be completed offline. Although it introduces additional time complexity, we believe that the performance improvement achieved through this method is well worth it.

6 Conclusion

In this paper, to fully exploit the information in the dataset, we proposed a data-centric knowledge tracing paradigm, LPDG, that aimed to generate interactive data between students and exercises by revealing their learning patterns and facilitating the generation of ideal training data. Specifically, the framework mainly consists of two modules: a learning patterns-guided exercise sequence regenerator, which was mainly used to capture students' potential learning patterns in the original dataset and generated new exercise sequence data with these learning patterns. Moreover, a learning pattern-guided pseudo-label generator simulated the evolution of knowledge states in the sequence interaction data and generate the student response data for new exercise sequence data. In the future we will prioritize validating educational consistency, optimizing generation efficiency, and enabling real-time deployment.

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